

Dildar Ali

Postdoctoral Research Candidate — Combinatorial & Submodular Optimization

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Research Interests

Submodular & Bisubmodular Function Optimization · Approximation Algorithms · Combinatorial Optimization · Influence Maximization · Regret Minimization · Algorithmic Data Management · Fairness in Multi-Agent Systems

About Me

I work on the design and theoretical analysis of approximation algorithms for NP-hard combinatorial optimization problems, primarily through the lens of submodular and bisubmodular function maximization, with applications to influence maximization, regret minimization, and algorithmic data management. I completed my Ph.D. in Computer Science and Engineering at the Indian Institute of Technology Jammu, advised by [Prof. Suman Banerjee](#) and [Prof. Yamuna Prasad](#). I am now seeking a postdoctoral position to extend this line of work toward scalable, provably-guaranteed algorithms for large-scale combinatorial and data-driven decision problems.

Education

Ph.D. in Computer Science and Engineering *Indian Institute of Technology (IIT) Jammu* Jan 2022 – Apr 2026

- Thesis: *Influential Slot Selection and Regret Minimization in Billboard Advertisement*
- Supervisor: Prof. Suman Banerjee · Co-Supervisor: Prof. Yamuna Prasad, IIT Jammu, India

M.Tech in Computer Science and Engineering *Aliah University, India* Jul 2018 – Jun 2020

- CGPA: 9.16/10 (University 2nd Rank Holder)
- Thesis: *A Deep Learning Approach to Improve Effectiveness and Efficiency for Time Series Prediction*

Research Contributions

- Developed approximation algorithms for NP-hard combinatorial optimization problems based on submodular and bisubmodular optimization with provable guarantees.
- Designed scalable algorithms for influence maximization and regret minimization under spatial, budget, and fairness constraints.
- Proposed efficient optimization techniques, including Pruned Submodularity Graphs, to accelerate large-scale submodular optimization.
- Extended optimization frameworks to multi-product, multi-channel, and multi-agent decision-making problems with theoretical analysis.
- Authored 23 peer-reviewed publications (6 journal articles, 17 conference papers) at venues including AAAI, PAKDD, DEXA, SEA, ADBIS, WISE, ACM SAC, ICAART, and EUMAS.

Experience

Teaching Assistant *Indian Institute of Technology Jammu* *Jammu, India*
Jan 2022 – Apr 2026

- Machine Learning (CSL774) — Instructor: Prof. Shaifu Gupta
- Database Management System (CSL362) — Instructor: Prof. Suman Banerjee
- Design and Analysis of Algorithms (CSC006P1M) — Instructor: Prof. Harkeerat Kaur
- Theory of Computation (CSL019U3M) — Instructor: Prof. Suman Banerjee
- Discrete Mathematical Structures (CS-203) — Instructor: Prof. Sumit Kumar Pandey
- Data Structures and Algorithms (CS-1-02) — Instructor: Prof. Suman Banerjee

Project Associate *Indian Institute of Technology (IIT) Bhilai* *Chhattisgarh, India*
Nov 2021 – Dec 2021

- Project: *Building Trust on Computing Platform and Training of Secure Coding of Security Chips*, funded by MeitY, Government of India
- PI: Prof. Dhiman Saha, IIT Bhilai, India

Publications

23 peer-reviewed publications: 6 journal articles, 17 conference papers, including AAAI, PAKDD ($\times 2$), DEXA ($\times 2$), ADBIS, WISE, SEA, ADC ($\times 2$), ACM SAC, ICAART, EUMAS, ADMA, and IKDD CODS.

Journal Articles

- [J6] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Joint Slot and Tag Selection in Billboard Advertising: A Bi-Submodular Maximization Approach.” [Submitted, **Knowledge and Information Systems**]
- [J5] Dildar Ali, Suman Banerjee, Rajibul Islam. “Trajectory-Driven Multi-Product Influence Maximization in Billboard Advertising.” [Under Revision, **Journal of Combinatorial Optimization**]
- [J4] Poonam Sharma, Dildar Ali, Suman Banerjee. “A Regret-Aware Framework for Effective Social Media Advertising.” **International Journal of Data Science and Analytics**. [Link](#) 
- [J3] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Minimizing Regret in Billboard Advertisement under Zonal Influence Constraint.” **Knowledge and Information Systems (KAIS)**. [Link](#) 
- [J2] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Billboard Slot Selection using Spatial Clustering and Pruned Submodularity Graph.” **International Journal of Data Science and Engineering**, 1–22. [Link](#) 
- [J1] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Toward Regret-Free Slot Allocation in Billboard Advertisement.” **International Journal of Data Science and Analytics** (2024): 1–25. [Link](#) 

Conference Papers

- [C17] Dildar Ali, Abishek Salaria, Ansh Jasrotia, Suman Banerjee, Yamuna Prasad. “Tag-specific Regret Minimization Problem in Outdoor Advertising.” [Submitted, **ACM CODS 2026**] [Link](#) 
- [C16] Dildar Ali, Ansh Jasrotia, Abishek Salaria, Suman Banerjee. “Approximation Algorithms for Budget Splitting in Multi-Channel Influence Maximization.” The 24th **International Symposium on Experimental Algorithms (SEA 2026)**. [Link](#) 
- [C15] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Approximately Bisubmodular Regret Minimization in Billboard and Social Media Advertising.” The 30th **Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)**. [Link](#) 
- [C14] Aditi Anand, Dildar Ali, Suman Banerjee. “Fairness Driven Multi-Agent Path Finding Problem.” The 18th **International Conference on Agents and Artificial Intelligence (ICAART 2026)**. [Link](#) 
- [C13] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Balanced Popularity in Multi-Product Billboard Advertisement.” The 36th **Australasian Database Conference (ADC 2025)**. [Link](#) 
- [C12] Dildar Ali, Rajibul Islam, Suman Banerjee. “Multi-product Influence Maximization in Billboard Advertisement.” The 13th **ACM IKDD International Conference on Data Science (CODS 2025)**. [Link](#) 
- [C11] Aditi Anand, Suman Banerjee, Dildar Ali. “Scalable Submodular Policy Optimization via Pruned Submodularity Graph.” [Accepted, **EUMAS 2025**] arXiv:2507.13834. [Link](#) 
- [C10] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Group Trip Planning Query Problem with Multimodal Journey.” The 36th **International Conference on Database and Expert Systems Applications (DEXA 2025)**. [Link](#) 
- [C9] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Slot and Tag Selection in Billboard Advertisement.” The 36th **International Conference on Database and Expert Systems Applications (DEXA 2025)**. [Link](#) 
- [C8] P. Sharma, Dildar Ali, Suman Banerjee. “Minimizing Regret in Social Media Advertisement.” The 12th **IEEE International Conference on Big Data (BigData 2024)**, pp. 473–478. [Link](#) 
- [C7] Dildar Ali, Suman Banerjee, Shweta Jain, Yamuna Prasad. “Fairness Driven Slot Allocation Problem in Billboard Advertisement.” The 29th **Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2025)**, LNCS vol. 15871. [Link](#) 
- [C6] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Multi-Slot Tag Assignment Problem in Billboard

Advertisement.” The 35th **Australasian Database Conference (ADC 2024)**, pp. 3–16. [Link](#)

- [C5] Dildar Ali, Harishchandra Kumar, Suman Banerjee, Yamuna Prasad. “An Effective Tag Assignment Approach for Billboard Advertisement.” The 25th **International Conference on Web Information Systems Engineering (WISE 2024)**, pp. 159–173. [Link](#)
- [C4] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Billboard Slot Selection Under Zonal Influence Constraint.” The 28th **European Conference on Advances in Databases and Information Systems (ADBIS 2024)**, pp. 93–106. [Link](#)
- [C3] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Regret Minimization in Billboard Advertisement under Zonal Influence Constraint.” The 39th **ACM/SIGAPP Symposium on Applied Computing (ACM SAC 2024)**, pp. 329–336. [Link](#)
- [C2] Dildar Ali, A. K. Bhagat, Suman Banerjee, Yamuna Prasad. “Efficient Algorithms for Regret Minimization in Billboard Advertisement” (Student Abstract). The 37th **AAAI Conference on Artificial Intelligence (AAAI 2023)**, 37(13), 16148–16149. [Link](#)
- [C1] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Billboard Slot Selection Using Pruned Submodularity Graph.” The 18th **International Conference on Advanced Data Mining and Applications (ADMA 2022)**. [Link](#)

Technical Skills

Programming Languages: C, C++, Java, Python

Tools & Platforms: L^AT_EX, Linux, Microsoft SQL Server, Oracle Database, NetworkX, Gurobi/CPLEX

References

Prof. Suman Banerjee [✉](#)

Assistant Professor

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Prof. Shweta Jain [✉](#)

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Prof. Yamuna Prasad [✉](#)

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Associate Professor

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